HENRY E. FALE
2918 S. 7TH. ST.
SHEBOYGAN, WI 53081
(414) 452-4172

H8SCOOP

FEBRUARY 1981
ISS#1102020281

BITS & PIECES

SUBSCRIPTION RENEWALS

STOP! Don't throw your envelope away yet. Look at the number opposite your name. It tells you the last number of your subscription. Compare this with the first two digits on the upper right hand side of the newsletter on ISS , which is the running issue number of the newsletter. This month's running issue is number 11, so if your number by your name is 12, next month will be your last issue.

For your information, the ISS is decoded as follows: The first 2 digits are the running issue number. As an example, this month ISS is 1102020281 which can be broken down to 11 02 02 02 81. The next two digits is the volume number, which increases with each year. Since this is the second year for H8SCOOP, the volume is 02. The next two digits (02 in this case) is the running issue this year, which is the 2nd for 1981. The last 4 numbers are the month and year of the present publication, which is 02 81, or February 1981.

Don't let your subscription laps, because I won't be sending out any warning notices to renew. When the last issue is sent, instead of a number appearing opposite your name, the word "LAST" will be printed by the computer to tell you this is your last issue of H8SCOOP unless you renew. Always check your envelope before you throw it away, it's the only way you will know when your subscription to H8SCOOP has ended.

If you noticed your subscription is running for a longer time than what you paid for, don't worry. I have extended the subscriptions to certain individuals for one reason or another.

As far as prices, the entire first 12 issue set is still available for the old prices, 12 issues for $12. This covers April 1980 to March 1981. Any issues after that, or renewals, will be as listed on the last page, $15 first class and $21 air mail overseas. As before, this will include the yearly index to H8SCOOP's, which may turn to every 6 months.

Also new, if you have noticed the heading, this publication is now copyrighted. It has been called to my attention that certain subscribing individuals are making a business out of xeroxing H8SCOOP and selling it at a reduced price.

Like I stated in a previous issue, you are only hurting yourself. I will now drive home that point. Anyone receiving xeroxed copies of H8SCOOP can get a lifetime subscription FREE by turning in the individual selling the copies. In turn, that individual will be taken off my mailing list, and turned in to the U.S. Government Copyright office for violation of copyright laws, which will be no fun. The individual giving me the info will not be disclosed to the persecuted. I hope this wakes some dishonest individuals up.

NEW PRINTER FOR H8SCOOP

I have just retired my selectric and added a DIABLO printer just before press time. It will take awhile to get used of the new printer and margin controls, so be patient. All bugs should be ironed out in a few months. This printer will allow me to obtain better print quality and formatting, once I learn how to use it. This issue is being prepared on PIE8, next issue I will try AUTOSCRIBE.

ON STRETCH-8

I understand from Del Stanton that some personal problems he has been involved in have delayed production and shipment of the STRETCH-8. Del assures me that he is still doing his best and slowly getting caught up on deliveries. He says by the end of January He hopes to be back on schedule.

It seems some of the individuals who have ordered or tried to check up on their order have been stifled by Del's Answering machine and Post office Box, getting no response. Del tells me he has sent checks back to all orders he now has in backlog, and will ship the STRECTH-8 to the individuals when they are ready, when that party can then either send Del a check or return the unit. This sounds fair to me.

I would caution buyers of anything to BEWARE of a POB and an answering machine with no definite way to get in touch with a person or party to verify anything.

INTERFACING COMPUTER TO THE WORLD

Many questionaires have been returned with interests indicated on interfacing their computers to the outside world. Many indicate a desire to be able to do this and want some direction.

For those not aware of my article series, or those who do not have a copy of my brief catalog abstract, send a SASE and request my catalog 3A and I will rush a copy off to you. I have printed project articles on "how to" interface with the computer, some being a 32 channel On/Off controller, A/D/A converter, Telephone circuit interface, I/O Selectric interface for disk system and more. If you have a parallel port for your computer, you're all set.

I will make a special offer this month to my readers to get things started. I will offer my 16/32 channel ON/OFF controller for $8 to those interested. The article tells you how to be able to control up to 32 devices from one 8 bit parallel port. The basic circuit costs about $20 or so to construct. To this you add relays, transistor drivers, solid state relays etc., to control the loads you desire. It is complete with many various driver circuits. It can be built on a small breadboard, wirewrap or PC type card. Normal cost is $12.

ON DON LONG'S CLOCK BOARD

Don called to tell me he will be taking orders and shipping about mid February for his clock-parallel I/O board as mentioned in past H8SCOOPS. I'll have more on this, along with an evaluation next month.

COMMENTS ON CONSTANT HEAD LOAD

Alden Finley writes "I have converted both of my H-17 drives to constant head loading and also employ my coast time PROLOGUE program on all bootable diskettes. . . To date, everything is going fine. The reduction in noise and relief from the irritation of having the disks spin on needlessly is a real joy. I have noticed no increase in disk errors or other indication of any wear." See September 1980 H8SCOOP for background on this.

SYSCOM REPORT ON TEIXEIRA'S SYSMOD

Alden also writes "I recently bought Jim Teixeira's SYSMOD and I've got to tell you that it is one peach of an improvement to HDOS. I used to use PIP all the time because I couldn't stand the disk grinding when I used CAT, etc. in HDOS, but now I stay in HDOS command mode and everything's a breeze. I haven't used PIP once since I installed SYSMOD. The biggest pain that Heath put in HDOS was DISMOUNT SY :. I swore everytime I had to type that; and now it's just D . DOCOM is a very convenient SUBMIT program with some extras and is a joy to use. All of the other helping features were very welcomed and I recommend SYSMOD to all H-8/H89 owners. . ."

FOLLOW UP ON TED BENGLEN'S H8 VOICE

Ted writes me the H-8 voice synthesizer is available to anyone interested. The price range is from $500 to $700 depending on options ordered. Ted also plans to sell the balance of the PC boards with complete documentation, diagnostic, demo program, and parts source list for about $50. Since Ted is only making 100 boards, anyone interested should place their orders as soon as possible to have a chance at getting the package.

Ted tells me the hardware and software have been redesigned to allow for voice inflection, which makes it easier to understand. The main part, the voice module from Vortax, normally sells for $400, but if you order Ted's package and mention to the company you are building Ted's project for the H8, you will be able to get it for $350.

Ted writes "I would like you to thank everyone for their interest and support, especially those who wrote to say they hoped the project would go, but were presently unable to make the deposit at that time..." See REQUESTS for more comments and address.

ON H9 KEYBOARD AND LOWER CASE MOD

From George Risk Industries, Inc. (GRI), GRI Plaza, Kimball, NE 69145 (308) 235-4645, A new keyboard retrofit for the H9, the H-75. I don't know the price. The bulletin sent to me by Wallace Izuo who saw it says "Owners of the H-9 Terminals can forget about scrapping this otherwise capable terminal with the addition of the GRI H75 replacement keyboard. While preserving the basic layout of the original keyboard, 8 added keys complete the full 128 character ASCII set, and both upper and lower case operation is standard. Minor modification is required to the H9 panel, but reliable KBM type keyswitches and rugged 2-shot molded keycaps guarantee years of dependable preformance from the H9. In addition, circuitry is provided on the keyboard for installation of an optional lower case character generator, completing the conversion. Retrofit takes less than 2 hours."

Wallace also sent me some information from Northwest Computer's lower case entry mod kit for the H9. Part of the bulletin says "Lower Case Entry is achieved by modifying Bits 5 and 6 of the output of the keyboard encoder chip. This method was used to hold down the cost. As a result, there are some minor BUGS that you will probably run across sooner or later. These bugs exist only in the lower case mode (SHIFT LOCK key up) and are associated only with the use of the SHIFT key. A SHIFT and a SPACE generate a ZERO and conversly, a SHIFT and a ZERO generate a SPACE. A SHIFT and a RUBOUT generates an UNDERSCORE. A SHIFT and a RETURN is not a RETURN. In addition to these, left and right brace, vertical bar and tilde cannot be generated from the keyboard."

"If you find that these deficiencies are unreasonable, you may return your Lower Case Entry within 30 days for a full refund."

Wally adds "I don't consider these deficiencies to be UNREASONABLE but if you are a fairly good typist, you WILL make errors (When I am using an ASCII typesphere on my Selectric and manually type, I get the greater/less than symbol sometimes when I type capital letters holding shift key down and run into period or comma. Minor, but SOMETIMES irritating.) Requires "conscious" typing which slows you down.

ON RATFOR DISK

Dave Heine writes in "I have been playing around with the RATFOR disk I got from you last month, and have discovered a few things about it. First, it is essential that you have a copy of Kernighan and Ritchie's book, "SOFTWARE TOOLS", to understand RATFOR. The version on your disk was derived from the C Language version of the source (RAT.C, RATDEF.C, and PUTLINE.C files.) I don't yet have a C complier, but intend to get one soon. Therefore I was not able to check whether I could compile the RAT.ABS file the same as the one on the disk. One problem with the RAT.ABS version supplied I noted is that it generates the C language run-time profile which tells you the number of times a function was called and how much time was spent in that function. This output is included in the output file and would have to be editted out before calling FORTRAN.

"There are a number of files on the disk with the file extension .RAT. These are the author's preliminary cut at coding up a RATFOR version as shown in the book mentioned above. To demonstrate the working of RAT.ABS you can enter:

```
  RAT  SY1:JUNK.  SY1:TEST.FOR
```

"This will process three of the .RAT files and produce a file called TEST.FOR. The FORTRAN code output looks strange because most of the blanks are omitted, which FORTRAN doesn't need, but make for readable listings. At any rate, your disk provides a lot of examples of C language code and also RATFOR code."

FEEDBACK ON CP/M EDITORIAL

James Quinn writes in "I've read the CP/M handbook and felt it helped me some with understanding HDOS and quite a few things I thought were very similar. I will buy CP/M but the only reason is to get alot of the programs that are available and hopefully I will be able to put them into HDOS via a program, and plan to use HDOS as my main operating system."

From Henry Beechhold, "I have to take issue with you on CP/M. Even though I have HDOS and a lot of HDOS software, I find that I use CP/M most of the time. Preparing new disks for use, for example, takes about a minute from start to finish! Compare that with the rigamarole you have to go through with HDOS. And the CP/M disks are bootable. Reconfiguring CP/M is likewise very simple--I play around with different baud rates, etc., with no problems. The CP/M RECONFIG file handles this nicely. CP/M's PIP is a very powerful utility, providing a number of switches and toggles. The EDitor is about on par with HDOS, which is to say that it's nothing speical. In CP/M, I use VEDIT exclusively. In HDOS, I use Jim Teixeira's EDGE, with is a vast improvement over EDIT. I don't do much assembly language work on my Heath system. . ."

TRS-80 SPEECH FOR H8/H89?

An article in December 1980 Microcomputing, Page 120 deals with "Speech Synthesis for the 6800". It tells how to use the speech synthesizer made by the Vortax division of the Federal Screw Works for the TRS-80 computers, and interface it to the SWTP 6800 computer. I believe if you have access to a parallel port for the H8 or H89 and have access to a TRS-80 synthesizer, you could make your computer talk.

The signals you will need are DATA BIT's 0 through 7, ADDRESS BITS 5 through 15, ground, and WRITE. The WRITE could be another signal set by software on the parallel port, such as DATA READY or something which can be set. You'd have to experiment on that, I just wanted to pass along the information.

ADDRESS CHANGE FOR DILITHIUM PRESS

Any books ordered from dilithium Press should be ordered from the following address, as there has been a change.

dilithium Press
POB 606
Beaverton, OR 97075

NEW STUFF

The Paper Tiger 460 has arrived. This is supposed to be a high density dot matrix printer with output good enough for word processing and graphics. It boasts a 24 X 9 dot matrix with overlapping horizontal and vertical dot formation, thus the high quality letters. In the graphics mode 84 X 84 dots per inch are possible. It prints at up to 9600 BAUD, and has many additional features, including proportional or mono spacing.

Some comments on Heath's PASCAL for $295 by Softech. Ralph Munroe reports the packaging and documentation is superb! There are some minor bugs in it though. If you have SYSTEM 2 disk, the compiler, there will be problems as Softech had problems with the copying. SYSTEM 2A will be sent to you if you write and/or send in your warantee card, which solves those problems.

Ralph also tells me there is a bug in the dumping of files to the H14, which will cause the H14 to hang up when copying files to it. Ralph called Softech, and got excellent results right away. According to Ralph, they have the makings to be in the GOOD GUY'S column.

Interface is via RS 232C interface. Line spacing is selectable at 6 or 8 lines per inch, printing characters are full ASCII upper and lower case. It looks like a low cost high speed word processing printer is finally available. Of course, if you have been following my comments in H8SCOOP, you already knew this was about to happen.

The basic model goes for $1295, and with the graphics option, $1394. For more info, contact your Integral Data Systems rep, or write Integral Data Systems, Inc., Milford, NH 03055.

I noticed the January 1981 issue of BYTE, CALIFORNIA DIGITAL, POB 3097 B, Torrance, CA 90503 (800) 421-5041 had the 460G, that's the printer with graphics option for $1150.

From Data Compass, POB 6259, Anaheim, CA 92806 (714) 630-7450, comes a COST-EFFECTIVE ALTERNATIVE TO THE H-47. They have available a solution to adding on mass storage, in configurations ranging from single headed single drive, single density to double sided, double headed, doubl density, giving a maximum of 3.2 Meg bytes in a single rack mounted enclosure.

The Remex "Intelligent" Model 4810 Master and Model 4820 Slave 8" floppy drives; double-sided, double-density, with interface cable for use with the WH-8-47 disk controller is $2,080 +$15 shipping with 90 day warranty. The Remex 4810, 4820 technical manual set is $40. Cabinets are also available.

This is the same drive the Heathkit version uses, and can be used with the H8 or H89 with appropriate controller interfaces (see Heathkit Catalog). It uses the standard HDOS disks. They are also selling the 4810 master for $1245 and the 4820 slave for $733, and have a 30 to 45 day delivery. For more information, contact Data Compass.

From SOFTWARE MART, 24092 Pandora St., El Toro, CA 92630 (800) 854-7115, Verbetim Mini-disks, 10 for $24.50. They accept VISA and MC. Their Source Mailbox: TCU155 and MicroNET Electronic Mail: 70341, 103.

APU-H ARITHMETIC PROCESSOR CARD for the H8 for $389 from CCM, Inc, POB 2308 Reston, VA 22091, is a high preformance processor card that greatly enhances the computational capability of the H8. Built around the AMD9511A arithmetic processor IC, it features 16 and 32 bit fixed point and floating point arithmetic.

The APU-H is easy to use and accessible as an I/O port by the H8. Contributing to the ease of use and versatility of the APU-H is a software package which replaces the standard BH EXT BASIC math routines with APU-H routines. It can offer substantial preformance gains (2 to 10 times faster) for many programs that are principally computational in nature.

UPDATE AND INFO FROM TRIONYX ELECTRONICS

Bill Perry from Trionyx writes me about the M-H8 64K dynamic memory board for the H8, which has proven to be a very popular and reliable item in 1980. Bill says not to worry about this board not being compatible with anything. "Heath and other compaines are introducing a wide variety of new products for the H8 computer. The M-H8 memory board can be made to operate with any new product developed for the H8 computer. We will supply design modifications, as required, to insure that the M-H8 will operate with these new products. The M-H8 will never become obsolete. We guarantee this - absolutely.

"The following is a list of design modifications currently under development for the M-H8 memory:

01-General Design Improvement
02-General Design Improvement
03-Z80 CPU adaptation
04-Zero Address Extension
05-4 MHz Operation

"These design modifications will be sent to those who are on our mailing list as having requested them, as they become available. These modifications are intended to be installed by the user. However, Trionyx will install and test any or all current modifications for a total fee of $25 on any M-H8 board sent to us for this purpose." Bill also writes this board WILL work with the new Heath H8 Color graphics board; the board draws less than 100 mA from the H8 18V power supply. Bill also tells me the board will work with the DG Z80 CPU board with the Z80 mod installed.

An update on the prices follows. For more information or for ordering, write Trionyx Electronics, POB 5131, Santa Ana, CA 92704.

Assembled no IC sockets	Kit with IC sockets	SIZE
$500	$415	64K
440	360	48K
380	305	32K
320	250	16K

16K memory expansion kit includes chips, sockets and capacitors--$60

Set of 8 Tested Memory Chips--$50

64K kit without memory chips--$225
64K Assembled without memory chips,
but sockets intalled--$300

Printed Circuit Board with Doc.--$50

From Micro Architect, INC, 96 Dothan St, Arlington, MA 02174 comes MBASIC HDOS software such as DATA BASE MANAGER, A/R, WORD PROCESSOR, MAILING LIST, INVENTORY, GL,A/R,A/P, and PAYROLL. For more information, send a SASE (28 cents) to the above address.

Jade Computer Products, 4901 W. Rosecrans, Hawthorne, CA 90250 (213) 973-7707 has the CAT MODEM for $139 plux tax and $2.50 shipping, and the D-CAT (Direct connect) for $179 plus tax and $2.50 shipping.

ABM Products, 631 "B" St, San Diego, CA 92101 has 10 BASF mini-disks for $24.

JDR MICRODEVICES, INC. 1101 South Winchester Blvd., San Jose, CA 95128 (800) 538-5000 has 200 ns 4116 dynamic memory chips for $29.95 plus tax and $2.00 shipping. These are the fast chips, and will work 8080 CPU or Z80.

Heathkit will soon be coming out with a mod to convert the H19 to an H88 (H89 minus disk drive) for under $500.

Walt Bilofsky of the Software Toolworks has some new software for sale which I will elaborate on next month. Briefly, an excellent video game for H19 and H89, INVADERS for $19.95. Also a full H89 graphics chessboard display for the MYCHESS program, with new self-play mode for same price of $34.95 or $40 for an update if you have purchased MYCHESS already.

Walt also is making most of his software available for Org 0 CP/M. More next month.

BIBLIOGRAPHY UPDATE

The January 1981 issue of Microcomputing, page 130 carried an article by Edgar Howell on "ENHANCING H8 BASIC". Written for tape EXT BH BASIC version 10.01.01, it allows you to read files created by the text editor.

REQUESTS

John Croft, 28016 Green Oaks Dr, Eugene OR 97402 writes in asking if the H8 system should be kept running 24 hours a day, or turned off when thru using? My thoughs are if you're using it, run it, if not, shut if off! There is usually a trade off point to consider. Component failure often occurs when power is first applied due to power surges through the equipment. The strain is usually more at this point than when left running. However, component failure is also related to time in operation, so it looks to me like six of one or half a dozen of the other. Anyone having any additional comments on this, I welcome your feedback.

Ted Benglen II, 822 E. Country Rd. 30, Ft. Collins CO 80525 writes "I would like to know if there is enough interest in a few skilled H-8 owners and all Users in general to form an H-8 development group. We would deal primarily in hardware development and related software support. It has occurred to me, there are probably several people working on the same things, and collectively, we could accomplish a lot more in a shorter time. I have some basic thoughts along these lines, and would like to hear from anyone interested along with their ideas.

"I believe the H8SCOOP presently is the only dependable line of H8 user communication. Keep it comming."

Let it be known, that this was one of my main objectives to starting this newsletter, to be a central exchange of ideas and to help others get in touch with those working on similar projects. That's why the REQUEST and WHO'S WHO columns have been included in H8SCOOP from the start. This was also one reason for the recent questionaire, to find who was doing what, and who wanted to know what.

I can print only what is given to me. As I said earlier, send your ideas, projects, questions, and requests in to me, and you can be sure, they will be printed in a future issue! I will continue to do all I can to make this information available to all readers. As soon as I weed through all the questionaires, I will start printing who is doing what with their computer, and things they would like the computer to be doing. Maybe this way, we can get more going and not re-invent the wheel.

I also know lot of HUG'S are doing good things and thus my call for HUG interaction in JANUARY H8SCOOP. I have received a few HUG newsletters, keep it coming HUGS!

Gary Baldus, 4228 Spruce St, Phila, PA 19104, is interested in organizing a Philly Hug. If anybody in this area is interested, contact Gary.

OFFERS

I make all software available on an as-is basis, in other words, no money back policy if not satisfied, mainly because it's too easy to copy disks. You take your chances. At my prices, you really can't go wrong. I try to get software out as fast as possible so you can get the newest stuff fast, and as thus, don't get a chance to carefully screen and test everything. To do so, would delay all software at least 3 to 6 months. I do the best I can and sell without assuming responsibility on the accuracy or runability of a program or disk. I don't think I've let anyone down in the past and will continue to do my best not to.

From now on, this column with past offers will be printed only every two or three months to allow more room for other goodies.

The following are past offers made by H8SCOOP, which are and will continue to be made available to those interested readers.

TINY PASCAL	$10
SEPT DISK--FILE DUMPS & UTILITY	$ 7
NPS CP/M MICRO COBOL SET(2)	$20
NPS CP/M MICRO COBOL SAMPLE TEST PGMS	$ 5
RATFOR TRANSLATOR	$ 7
NOV. DISK--PICTURES, LABELS, WORD PRO	$15
QUIKMAIL MAILING LIST PROGRAM	$15
C SOFTWARE STUFF DISK	$10
DEC. DISK--PICTURES & UTILITIES	$10

H19 GRAPHICS PROGRAMS DISK--1/81 $10
DOUBLE LA-36 DEC SPEED ARTICLE $ 6

This month's offer for $10 is for a disk with some really neat utilities, a PLANT program, SWARMS (a game to stop the killer bees),and a STARWARS.BAS which the computer will play by itself if you desire. Home Plants requires an H89 or H8 with H19 or H9, 40K memory, single floppy and MBASIC. It tells you what plants grow in various environments and the moisture, sunlight etc. required for best results.

SWARMS requires an H89 or H8 with H19, 32K, single floppy and Extended Benten Harbor Basic. It is a very complicated game taking 15 or so hours to win, to stop the killer bees. Both of the above were submitted by Haywood Nichols Jr.

The utilities include COPYC, a copy program which copies files contiguiously. This program in itself is worth the money. It will copy a program in sector and track order on a disk. The advantage? When loading a program or data from disk it will load faster with less disk activity, which means less wear and tear on your disk drive.

Also with the utilities are a new SUBMIT which uses two CTRL "D"'s to terminate (thus you can use it on FLAGS and other programs which require CTRL D), and a modified PATCH which doesn't require ID, and UNLOCK, to unlock all flags on a disk for your modifying.

PROGRAMMING GOODIES

There's all kinds of neat disk dump programs available, but what if you want to find out where things are stored in memory? Here is a short simple program I wrote from an idea by John Welch to allow you to print out the contents of any memory location(s), as many as you want. You can print to a hard copy device, or to console only. If you print to hardcopy, it is also printed to console automatically.

This version is for MBASIC. BH EXTended BASIC can be used by changing line 3 to OPEN A$ FOR WRITE AS FILE 1. Only printable characters are printed, non printable characters, or characters which may be escape codes to screw up an H19 are printed as a dot (.). It is set for 70 characters/line with the first number being the memory location start in decimal. You can find lot of useful stuff by runing this program. The MBASIC version runs much faster than the BH version. Remember in MBASIC to load hard copy device driver before running MBASIC.

```
1 REM ** MEMORY EXAMINER ROUTINE BY
  HENRY FALE FOR H8SCOOP
2 PRINT:LINE INPUT"ENTER COPY DEVICE IF
  APPLICABLE, EX AT:   ";A$
3 OPEN "O",1,A$
5 INPUT "ENTER LOW MEMORY LIMIT TO BE
  EXAMINED   ";X1
8 INPUT "ENTER HIGH MEMORY LIMIT TO BE
  EXAMINED   ";X2
10 FOR X=X1 TO X2 STEP 70
11 PRINT:PRINT X;TAB(7);:PRINT #1,:
   PRINT #1,X;TAB(7);
12 FOR Y=0 TO 69
14 C$=CHR$(PEEK(X+Y))
15 IF C$<CHR$(32) OR C$>"z" THEN C$="."
16 PRINT C$;:PRINT #1,C$;
20 NEXT Y
30 NEXT X
35 CLOSE #1
37 PRINT
40 END
```

If you do not have hardcopy, eliminate all Print 1, statements. By runing this program, you find locations 1000 decimal to 5100 blank, which are reserved and presently not used. Locations 5601 to 5811 contain the present SY0: disk directory. Locations 5900 to 5951 contain the SY1: disk name if mounted.

You can run through your entire memory with this and discover where lots of things are hidden.

Another good short program submitted by Professor Grant Gustafson is a FREE SECTOR POOL routine which from BASIC or MBASIC will return the number of free sectors remaining on a disk, good for knowing if the program or data you are working with may be getting too large to dump on the disk!

```
1 F=0:J=0
2 J=PEEK(5120+J)
3 IF J=0 THEN 6
4 F=F+2
5 GOTO 2
6 PRINT F;" SECTORS FREE."
7 END
```

You can keep this as an ASCII TEXT file on SY0:, and merge it with any existing BASIC or MBASIC program. To keep it from killing programs on a merge, start the numbering at 65490 instead of run, and do a RUN 65490 to use it.

The MERGE technique is useful for answering questions like "will I lose part of my old file if I UNSAVE (KILL) it and then rewrite it with the one in memory?"

TECHNICAL FORUM

ON RADIO FREQUENCY INTERFERRENCE

Now that the H8 color board has been introduced, I wonded what the effect of RFI would have on the nearby Color TV receiver. My H8 really messes up my color receiver, located far away, on a different floor and room. I saw it demonstrated recently, and maybe because of direct video, there was no intereferrence on the color monitor it was attatched to.

As you may be aware, the FCC has clamped down on computers and their associated devices producing RFI. The H8 is being re-worked to eliminate this. My experience shows me the majority of this RFI is being radiated from the H8 and cables into the atmosphere. This is a problem only shielding can solve. In case you did not know, the sides of the H8 are a plastic, not metal, which does not help at all to limit the RFI.

I print some instructions from Stu Gurske, 2809 Dewey Ct, Middleton, WI 53564, who is a Ham operator. Stu writes these instructions for cutting down RFI on the STRETCH 8 he built, but the general idea should apply equally well to the H8, as I had similar ideas for some time now (but no time to try it).

First paint the inside surface of both sides with contact cement. Cut some aluminum wrap to the size of the sides and coat the non shiny side with contact cement. Now place this aluminum foil on the inside sides of the cabinet. Next remove the black paint from the lips of the bottom of the enclosure. Be sure the aluminized sides make contact with the metal bottom and top of the H8. This should shield out much of the RF interference. — See pg 8

CLASSIFIEDS

FOR SALE--H8-2 3 channel parallel board for H8 computer. 1 channel is not working. Make offer. Don Long, 9319 Willowview, Houston, TX 77080.

FOR SALE--RO 33 Teletypes (6 available). All have been tested using the H8-4 serial card. Cheap reliable hard copy at $210 each. You pay shipping from Salem, OR. Call Terry Hagel at 503-581-7175.

FOR SALE--H9 Video Terminal $150. Excellent Condition, New Key Caps. Will trade for 12K/16K memory board. Tom Roche, 29W476 Blackthorn Ln, Warrenville, IL 60555 (312) 393-2023.

FOR SALE--IBM model 735 I/O selectric PICA typewriter, about 1 year old. Comes complete with interface for H8 parallel card (which I'll sell if you don't have one), fabric and cloth ribbons, device driver disk, schematics, maintenance and operation manuals, source code, and complete electronics package. Just plug it in and away you go. Will work with AUTOSCRIBE. Excellent condition. This is not a re-furbished unit, but a new IBM unit with few hours on it. Will double as a stantard typewriter. This is the machine H8SCOOP has been printed on the past year. $1000, you pick up. Contact H8SCOOP.

FOR SALE--Integral Data systems IP-125 PRINTER. With Printer Control Options. 64 to 132 Column. Full ASCII Upper and Lower Case, 256 Byte Buffer, RS232 Serial, up to 1200 BAUD. Friction Feed. Use in H8 with ATH-85 device driver. Modified H8-5 Serial Board included for Clear-to-Send Handshaking. Complete with H8-5 board and roll paper holder, $400. Ralph Bradburn, 300 Route 59, Tallman, NY 10982.

FOR SALE--Dual drive H17. Works perfectly, with 6ms step time $700. Will seperate drives in case $550; controller and cable, $150. Larry Bledsoe, MD, 1350 W. Bethune 803, Detroit, MI 48202 (313) 871-3759.

FOR SALE-- Available Immediately: H-8 Computer, H8-5 serial I/O board for cassette and CRT, H-9 CRT Terminal, 2 H8-8 8K memory boards, and a cassette player. Sale price is negotiable. Please call Dan Baumgartner at (312) 325-3487 after 6PM or on weekends.

FOR SALE--IBM Selectric I/O terminal. Not a hybird; 100% IBM, so they will service it. Comes working, interfaced to H89 for printing only, with HDOS driver, and source code, four type elements, 12 film and a cloth ribbon. $800. You pick up. Walt Bilofsky, 14478 Glorietta Drive, Sherman Oaks, CA 91423. (213) 986-4885.

WANTED--H-77 for H89. Call (714) 882-8072 collect if you are interested.

EDITORS MAILBOX

CONTROVERSY ON DUAL DENSITY 5 1/4

Since I have been receiving many letters on my comments on dual density I will use this column to make all aware of what's going on.

I have printed in the past that it is possible to go double density on the H17 with no hardware modifications. I have also printed this was about to take place with a target date around April 1981. It turns out this is not happening. So what gives? Did I make this all up? NO! I have had reliable sources for all my information. At this point I honestly don't know what's happening, it appears politics may be playing a major role in the setback.

First off, to defend my integrity. The manual supplied with the drive unit itself, states "Using MFM or MMFM encoding techniques, double density data storage to 249.4K bytes can be achieved on a single side of the diskette." So that takes care of the drive unit itself, it WILL handle double density data storage.

I don't like to print letters not addressed to me, but I will print part of a letter written by Barry Watzman, Product Line Manager of Heath Company, written May 23, 1980 to Mr. "X". "Higher Capacity 5" drives are under evaluation. The current controller will support double sided 96 tpi drives (400K/drive, 5"), with no changes. However, there are major software issues and some reliability problems to be resolved before such a system is offered."

Since it appears the controller will handle another drive to give quadrouple density it would seem logical that minimum changes would be required to adapt this to the present drives, since they are capable of handling double density. Maybe not quad density, or double sided, but we never asked for that, simply double single sided density as inexpensively as we can get it.

To top this off, an individual from a local HUG called to tell me Jim Blake appeared recently and stated double density drives are being developed at Heath, and we can expect to see them about April of 1981.

In a letter written by Steve Parker, Technical Consultant of Heath company to Mr. "Y" Dated Octover 28, 1980 "Although a double density, double sided 5" disk system is under consideration, there are no firm plans to begin engineering on it at this time. Should this product be developed, it will require new disk drives, as the current ones do not have double density or double sided capabilities. On the other hand, the current disk controller board has capabilities for both of these types of operation."

So who's talking double sided??! From this letter I infer the controller board will handle it, but the drives won't. Wanco tells me the drives will! If the drives will and the controller will, what's the problem. Like I said, I print only what I receive from reliable sources, and in this case, there appears to be mass confusion (or mass coverup) from some people at Heath!

So there you have it. Now the 8 inch drives are out, and suddenly nobody said anything or knows anything or will admit to anything definite being in the works. My opinion on all this is they may be holding off to sell the high priced 8 inch drives. All the energy seems to be in this direction. It is strange that of all the small micros; Heath, Apple, Pet, Radio Shack, and whoever else started after Heath and had only cassette storage while Heath had disk drives. Now they have all passed up Heath with not only disk drives, but many having double density, and here we sit with our single density.

Methinks something is not Kosher here, and we should continue pressuring Heath for double density drives.

ON H19 DRIVE HOUSING MOD

Mount a drive in the H19 at your own risk. Individuals have done it with no apparent problems, but the warning was given that there may be shielding problems causing trouble with the computer system. I print part of a letter here from Steve Parker on the subject. "We strongly recommend against an attempt to install a disk drive in the H-19 video terminal. Although we use a disk drive in the H-89 computer, many special engineering techniques were required to make this combination operate properly. Attempting to install a drive in the H-19 would almost surely cause severe technical problems which would adversely effect the entire computer system."

So that's the word from Heath, you'll have to decide yourself.

Henry

ADDENDUM TO H8 RFI ARTICLE: I omitted some important parts to this article and felt I better warn you. Extreem care must be taken to be sure no shorting of the motherboard or contacts will result, or PUFF! Use care to be certain that the aluminum on especially the right side will not come in contact with any of the H8 BUSS pins.

****

"HEATH" and "HEATHKIT" are registered trademarks of Heath Co.

MARCH issue deadline--February 13

Subscription Rates--$15/Year ($21 overseas AIRMAIL)

PHONE HOURS: 6PM-10PM Monday, Tuesday, Thursday, and Sunday. No collect calls